

HO CHI MINH TAN SON NHAT INTL, Viet Nam

WMO: 489000

Lat: 10.819N

Lon: 106.652E

Elev: 10

StdP: 101.20

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.1	21.2	14.9	10.6	26.5	16.2	11.5	27.0	6.9	27.6	6.1	27.6	1.8	0	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.6	35.8	25.7	34.9	25.7	34.1	25.7	28.1	31.5	27.7	31.1	27.4	30.9	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	29.4	27.0	22.7	29.4	26.5	22.1	29.0	90.2	31.6	88.4	31.7	87.2	31.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.6	6.0	DB	18.3	37.2	1.2	0.7	17.4	37.7	16.7	38.1	16.0	38.6	15.1	39.1	(4)
(5)			WB	16.0	29.0	1.4	0.8	15.1	29.6	14.2	30.1	13.5	30.6	12.5	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.0	28.0	29.3	30.2	29.8	28.8	28.3	28.3	28.1	27.8	27.6	26.9	(6)
(7)		DBStd	1.58	1.43	1.21	1.10	1.17	1.45	1.17	1.08	1.08	1.08	1.07	1.22	1.49	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6697	528	503	599	607	614	564	567	568	542	552	529	523	(10)
(11)		CDD18.3	3656	270	269	341	357	356	314	308	309	292	294	279	265	(11)
(12)		CDH23.3	38317	2569	2730	3892	4390	4173	3411	3107	3155	2855	2802	2723	2509	(12)
(13)		CDH26.7	15628	996	1144	1740	2122	1930	1387	1167	1168	1015	997	1027	934	(13)
(14)	Wind	WSAvg	2.8	2.4	2.9	3.4	3.3	2.8	3.0	3.3	3.4	2.8	2.2	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	1752	15	7	30	64	194	236	253	257	282	260	122	33	(15)
(16)		PrecMax	2108	90	57	119	228	308	385	554	422	518	390	315	97	(16)
(17)		PrecMin	1375	0	0	3	5	32	64	110	77	154	142	2	1	(17)
(18)		PrecStd	191	22	13	33	56	72	86	100	89	80	73	78	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.0	36.2	36.4	36.8	35.1	34.2	34.1	34.1	33.9	34.0	33.9	(19)
(20)			MCWB	23.8	23.5	24.4	25.5	26.4	26.8	26.7	26.6	26.6	25.6	25.0	24.0	(20)
(21)		2%	DB	33.1	34.0	35.0	35.5	35.7	34.1	33.2	33.2	33.1	32.9	33.0	32.9	(21)
(22)			MCWB	23.9	23.7	24.9	25.8	26.4	26.8	26.5	26.5	26.6	25.9	25.3	24.4	(22)
(23)		5%	DB	32.1	33.1	34.1	34.8	34.3	33.2	32.8	32.7	32.2	32.0	32.1	32.0	(23)
(24)			MCWB	23.8	23.8	24.8	25.9	26.6	26.6	26.4	26.4	26.4	25.9	25.3	24.3	(24)
(25)		10%	DB	31.2	32.1	33.1	33.9	33.3	32.2	31.9	31.9	31.3	31.2	31.4	31.1	(25)
(26)			MCWB	23.7	23.7	24.7	25.8	26.6	26.5	26.3	26.3	26.3	26.0	25.4	24.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.4	27.2	28.1	28.5	28.5	28.2	28.2	28.0	27.8	27.5	26.8	(27)
(28)			MCDWB	30.4	31.2	32.0	32.3	32.1	31.6	31.1	31.4	31.4	30.7	30.7	30.1	(28)
(29)		2%	WB	25.9	25.7	26.6	27.6	27.9	27.9	27.6	27.5	27.5	27.2	26.8	26.2	(29)
(30)			MCDWB	29.7	30.5	31.3	31.6	31.4	31.2	30.7	30.9	30.8	30.3	30.0	29.6	(30)
(31)		5%	WB	25.3	25.2	26.2	27.1	27.5	27.5	27.2	27.1	27.1	26.8	26.4	25.8	(31)
(32)			MCDWB	29.3	29.9	30.8	31.0	31.2	30.9	30.4	30.5	30.4	29.9	29.6	29.2	(32)
(33)		10%	WB	24.7	24.7	25.7	26.7	27.1	27.1	26.8	26.7	26.6	26.5	26.0	25.3	(33)
(34)			MCDWB	28.8	29.3	30.2	30.7	30.9	30.5	30.0	29.9	29.7	29.4	29.2	28.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	9.3	8.6	8.4	8.0	7.8	7.6	7.5	7.7	8.2	8.8	(35)
(36)			MCDBR	10.1	10.4	9.8	9.1	9.0	8.8	8.6	8.5	8.4	8.5	8.9	9.5	(36)
(37)			MCWBR	3.3	3.2	2.8	2.6	2.9	3.2	3.3	3.1	3.2	3.2	3.2	3.3	(37)
(38)		5% WB	MCDBR	8.8	9.2	8.8	8.4	8.4	8.1	7.8	7.6	7.6	7.7	8.1	8.3	(38)
(39)			MCWBR	3.3	3.1	2.8	2.9	3.3	3.5	3.5	3.3	3.4	3.4	3.3	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.488	0.513	0.551	0.530	0.461	0.452	0.452	0.455	0.460	0.495	0.461	0.461	(40)	
(41)		taud	2.157	2.078	2.012	2.093	2.335	2.382	2.372	2.356	2.331	2.219	2.314	2.276	(41)	
(42)		Ebn at Noon	819	813	788	797	838	836	839	848	852	818	839	835	(42)	
(43)		Edh at Noon	151	168	183	167	128	121	123	127	131	144	128	132	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.13	5.88	6.01	5.92	5.35	5.01	4.91	4.92	4.83	4.82	4.85	4.71	(44)	
(45)		RadStd	0.28	0.26	0.28	0.28	0.36	0.21	0.27	0.34	0.33	0.39	0.29	0.38	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.22	+0.38	N/A	N/A	+0.22	+0.28	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page